

# Solver-Aware Neural Dynamics

Interrogating Integrators and Basis Functions in Kolmogorov–Arnold Network ODEs

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## COURSE

**CSE 402: Numerical Analysis, Simulation & Modeling**

## PRESENTATION

Project Proposal Presentation

## DATE

17 August 2026

### Team Members:

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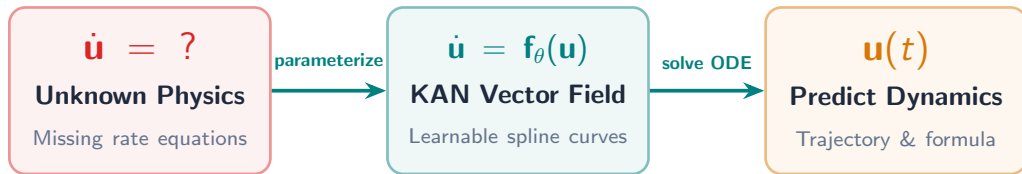
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# Base Paper: Learning the Missing Physics



**Base Paper:** Z. Koenig, J. Kim, Y. Deng, *KAN-ODEs: Kolmogorov–Arnold Network Ordinary Differential Equations for Learning Dynamical Systems and Hidden Physics*, CMAME / arXiv:2407.04192 (2024), MIT.

**Learn the physics,  
don't assume it.**

# Base Paper's Method: Three Key Decisions

## 1. Integrator

*Forward Integration Pass*

- **Tsit5** adaptive RK 5/4 solver
- Continuous-time forward state
- Grid-independent evaluation

## 2. Activation

*KAN Architecture*

- Learnable 1D edge curves
- **Gaussian RBF** basis
- Symbolic law discovery

## 3. Optimization

*Backward Pass & Training*

- **Adjoint Sensitivity** gradients
- **Adam** with  $L_1$  sparsity
- Memory-efficient backprop

**10× Fewer Epochs**

Lotka–Volterra:  $10^4$  vs  $10^5$  epochs

Parameters: **240** (KAN) vs 252 (MLP)

# Our Extension: Stress-Testing the Design Choices

The base paper fixed **Tsit5** + **Gaussian RBF**. We systematically interrogate both choices.

## 1. Solver Ablation

*maps to Integrator*

- Forward Euler ( $p = 1$ )
- Heun RK2 / Classical RK4
- Adaptive Dormand–Prince

**Metrics:** Loss · NFE · Drift

## 2. Basis Swap

*maps to Activation*

- Cubic B-splines ( $k = 3$ )
- Gaussian RBF (*baseline*)
- Lagrange & Newton Poly

**Metrics:** Accuracy · Params

## 3. Robustness

*new beyond the paper*

- Damped Pendulum dynamics
- SIR Epidemic dynamics
- Noise sweep ( $\sigma \leq 0.10$ )

**Metric:** Generalization

**Controlled Setup:** Adjoint gradients and Adam optimizer remain fixed, strictly isolating solver and basis effects.

# Expected Methodology: One Reproducible Pipeline



## Results Deliverable:

- Train/Extrapolation MSE · NFE per step
- Sensitivity to noise ( $\sigma \leq 0.1$ ) · Wall-clock speed

## CSE 402 Syllabus Convergence:

Euler/RK ODE integration · Spline & polynomial interpolation · Truncation error · Optimization